

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

DIGITAL TWIN OF A SAILBOAT
BACHELOR THESIS

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FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

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BACHELOR THESIS

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Abstract

Abstract in the English language (translation of the abstract in the Slovak language).

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Úvod

Cieľom tejto práce je poskytnúť študentom posledného ročníka bakalárskeho štúdia informatiky kostru práce v systéme LaTeX a ukážku užitočných príkazov, ktoré pri písaní práce môžu potrebovať. Začneme stručnou charakteristikou úvodu práce podľa smernice o záverečných prácach [?], ktorú uvádzame ako doslovný citát.

Úvod je prvou komplexnou informáciou o práci, jej celi, obsahu a štruktúre. Úvod sa vzťahuje na spracovanú tému konkrétne, obsahuje stručný a výstižný opis problematiky, charakterizuje stav poznania alebo praxe v oblasti, ktorá je predmetom školského diela a oboznamuje s významom, cieľmi a zámermi školského diela. Autor v úvode zdôrazňuje, prečo je práca dôležitá a prečo sa rozhodol spracovať danú tému. Úvod ako názov kapitoly sa nečísluje a jeho rozsah je spravidla 1 až 2 strany.

V nasledujúcej kapitole nájdete ukážku členenia kapitoly na menšie časti a v kapitole ?? nájdete príkazy na prácu s tabuľkami, obrázkami a matematickými výrazmi. V kapitole ?? uvádzame klasický text Lorem Ipsum a na koniec sa budeme venovať záležitostiam záveru bakalárskej práce.

Chapter 1

Theoretical Foundations

1.1 Digital Twins

A digital twin is a virtual model of a physical object or system. This model is then continuously updated based on real-world data from the physical object. In the context of this project, it is also updated based on environmental data in a simulated environment. Digital twins allow us to monitor the state of objects in real time, predict their behavior, and optimize their operation. In the case of a sailboat, a digital twin helps simulate and analyze its behavior in environmental conditions, such as different wind and water conditions. We can also see the behavior of a given sailboat digital twin with different sail settings. [5]

1.2 Fundamentals of Sailing Physics

Sailing is based on the effects of two different environments acting on the ship. The first environment is the air or wind, which affects the sails and other parts of the ship above the water at a given moment. The second environment is the water, which affects the ship through waves and underwater currents.

True and Apparent Wind

True wind is the so-called true wind that is measured relative to a stationary object on the water surface or on land. It is expressed by the speed TWS and the angle TWA. These quantities are constant and are not affected by the movements of bodies. They change only when the wind itself changes its speed or direction, not when the body itself that captures it changes its speed and direction. If a given ship is moving, its crew does not feel the true wind or the so-called apparent wind. This arises when the ship's vector is subtracted from the true wind vector. [1]

$$\text{Apparent Wind} = \text{True Wind} - \text{Boat Velocity}. \quad (1.1)$$

For example, when the wind blows at a speed of 16 km/h directly in front of the ship, and the ship goes at a speed of 7 km/h. So, the apparent wind will be 9 km/h because $16 - 7 = 9$. Therefore, apparent wind is used in almost all calculations instead of true wind because it takes into account the direction and speed of movement of the ship.

Lift and Drag

Once the apparent wind vector acting on the boat is known, the two principal aerodynamic forces on the sails, lift and drag, can be calculated. Lift arises whenever the sail meets the apparent wind at a non-zero angle of attack. In that situation, the airflow over the leeward side of the sail accelerates, and its pressure drops, while the windward side remains at higher pressure. The resulting pressure difference generates the lifting force. Lift can be divided into two components, a forward component that drives the boat ahead and a lateral component that heels the boat to the side. [2]

Drag, by contrast, is the force that acts on the boat in the direction of the airflow and slows its motion when the boat moves upwind. The smaller the drag, the more easily the boat can accelerate. When the boat is sailing with the wind, that drag becomes the dominant component of the force moving the vessel forward. [2]

$$\begin{aligned} F_{\text{lift,sail}} &= 0.5 \rho_{\text{air}} S C_L(\alpha) V_{\text{aw}}^2, \\ F_{\text{drag,sail}} &= 0.5 \rho_{\text{air}} S C_D(\alpha) V_{\text{aw}}^2, \end{aligned} \quad (1.2)$$

The factor $\frac{1}{2}$ is a constant that appears in the definition of dynamic pressure. The term ρ denotes the air density; denser (colder) air produces a larger force. The velocity V is the speed of the apparent wind and it is squared, so doubling the wind speed multiplies the force by four. The coefficient C represents the aerodynamics of the sail: C_L when we compute lift, C_D when we compute drag. Its value depends mainly on the angle of attack. Finally, S is the area of the sail, the larger the area exposed to the flow, the greater the resulting force.

The same quadratic law is later reused to estimate the forces on the rudder. By replacing the air density ρ_{air} with the water density ρ_{water} and the sail area S with the rudder area S_{rudder} , yields the hydrodynamic lift that acts on a deflected rudder blade. Multiplying this lift by the horizontal distance between the rudder's centre of pressure and the boat's centre of gravity provides the yawing moment that turns the vessel.

1.3 Photogrammetry

Photogrammetry is a measurement technique that reconstructs a three-dimensional model of a scene from video or overlapping photographs. The method is based on triangulation. An algorithm that detects identical image features in several views, solves for the camera poses, and computes the spatial coordinates of each feature point, from which it generates a dense 3D mesh. [6]

Modern photogrammetric workflows fall into three practical categories:

1. Image-based SfM/MVS pipelines (e.g., COLMAP, RealityCapture) rely solely on RGB imagery and achieve centimeter-level accuracy when the surface carries enough texture.
2. Hybrid image + LiDAR workflows merge passive photographs with active laser or depth scans, combining the color fidelity of photos with the scale-stable, texture-independent geometry of LiDAR—useful on glossy or uniformly colored surfaces that challenge feature matching.
3. Neural approaches such as Neural Radiance Fields (NeRF) and single-image 3-D networks learn a continuous radiance field or volumetric representation from the photos themselves; they excel at photorealistic renderings with sparse or imperfect image sets, though their metric precision still trails classical methods. [7]

Successful photogrammetric modeling depends not only on the software used to generate the 3D mesh but also, crucially, on the quality of the photo dataset itself. The images must overlap extensively, ideally by more than 70%, so that the algorithm can reliably triangulate feature points. When shooting, it is not enough to move the camera only from left to right; additional high and low angles are needed to prevent gaps that arise where the object’s surface curves out of view. [6]

Camera choice also matters. Wide-angle lenses introduce barrel distortion that can create holes or deformations in the final mesh, so such lenses must be calibrated either before the shoot or during processing. Finally, photographs contain no inherent scale information. Individual parts of the reconstructed sailboat may be proportionally correct, yet the entire mesh can end up larger or smaller than the real vessel. The scale ambiguity can be resolved by placing a reference object of known length in the scene while shooting or by merging the photo reconstruction with a LiDAR scan, which provides an absolute metric framework.

Even with careful shooting and a well-constructed dataset of photos, the resulting 3D model can still exhibit mesh defects. Accuracy depends not only on the software but also on the object’s surface characteristics. Smooth or uniformly colored areas provide too little texture for the algorithm to match features reliably, so holes appear

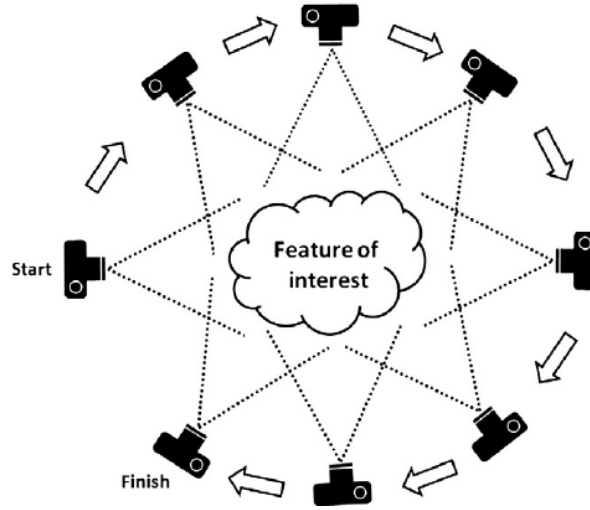


Figure 1.1: Schematic of photogrammetry image capture [8]

in the point cloud. Glossy or wet surfaces worsen the problem because reflections and specular highlights mislead the feature detector and generate “floating” points. The so-called “floating” points are spurious vertices that do not lie on the true surface. Instead, they end up hovering in front of the hull, behind it, or entirely outside the mesh boundary. Deep shadows and strong contrasts introduce another set of complications because shaded regions have low local contrast, reducing the number of identifiable points and producing noise or blank patches in the reconstruction. A further limitation is line-of-sight. Neither photographs nor LiDAR beams can capture geometry hidden beneath the deck. Occluded areas, therefore, reconstruct poorly or not at all, leaving distorted or self-intersecting faces that must be removed or patched in the post-processing phase. [6]

Applied to a sailboat, photogrammetry is especially valuable because it captures fine details of the hull and deck hardware that would be difficult or time-consuming to replicate by hand. The resulting digital model is, therefore, more accurate and visually convincing while requiring less manual modeling effort.

1.4 Overview of Software and Technologies

After the 3D model was created using photogrammetry, two key applications were used to turn it into a functional digital twin. Blender was first employed to clean and refine the 3D model of the hull and deck. The finished geometry was then imported into Unreal Engine 5, where the movement of the boat’s parts was animated, a surrounding environment with realistic weather and sea conditions was built, and the physics formulas governing speed and heading were implemented.

1.4.1 Blender

Blender is a free and open-source 3D computer graphics software tool that integrates modeling, texturing, sculpting, UV mapping, and animations. Within this thesis context, it was used to clean up and prepare the sailboat mesh and divide the model into parts that can then be individually imported into the Unreal Engine. It was used to find and seal all holes created using photogrammetry and to remove the redundant geometry. The model was then rotated to the correct X, Y and Z-axes, aligned to Unreal Engine's coordinate system, and scaled to the needed units. As the last step, it was then exported as an FBX file so it could be safely imported into the Unreal Engine in the form we wanted. For example, the hull will be mesh, but the sails will be imported as a skeletal mesh. [3]



Figure 1.2: A sample screenshot of fixed holes from photogrammetry

1.4.2 Unreal Engine

Unreal Engine is a 3D game engine used for everything from game development to filmmaking and simulations. Its functionality can be accessed through conventional C++ code or visual scripting in Blueprints. In this thesis, it was used to create a simulated marine environment, including waves, wind, sky, and dynamic lighting, into which the sailboat's digital twin was placed. [4]

To create the required marine environment in Unreal Engine, several plugins were enabled:

1. Landmass plugin was used to sculpt the coastline and seabed. Allowing for some ground to exist to test collisions.
2. Water plugin generated a dynamic ocean surface with Gerstner waves.
3. Buoyancy plugin computed the lift acting on the submerged parts of the hull each frame based on the pontons positioning and water depth.

After enabling these plugins, the cleaned FBX model of a sailboat was imported into the project. This model is separated into different parts and then added to an actor as meshes to create one object that we can manipulate and move. Afterward, Chaos Cloth was used to simulate fabric behavior and was applied to sails. If sails have the correct mesh composition, it will allow them to deform and simulate real fabric in a windy environment.



Figure 1.3: Prototype in Unreal Engine

Chapter 2

Mesh Preparation in Blender

Chapter 3

Real-Time Environment in Unreal Engine 5

Chapter 4

Physics Implementation

Záver

Na záver už len odporúčania k samotnej kapitole Záver v bakalárskej práci podľa smernice [?]: V závere je potrebné v stručnosti zhrnúť dosiahnuté výsledky vo vzťahu k stanoveným cieľom. Rozsah záveru je minimálne dve strany. Záver ako kapitola sa nečísluje.

Všimnite si správne písanie slovenských úvodzoviek okolo predchádzajúceho citátu, ktoré sme dosiahli príkazom \uv.

V informatických prácach niekedy býva záver kratší ako dve strany, ale stále by to mal byť rozumne dlhý text, v rozsahu aspoň jednej strany. Okrem dosiahnutých cieľov sa zvyknú rozoberať aj otvorené problémy a námety na ďalšiu prácu v oblasti.

Abstrakt, úvod a záver práce obsahujú podobné informácie. Abstrakt je kratší text, ktorý má pomôcť čitateľovi sa rozhodnúť, či vôbec prácu chce čítať. Úvod má umožniť zorientovať sa v práci skôr než ju začne čítať a záver sumarizuje najdôležitejšie veci po tom, ako prácu prečítal, môže sa teda viac zamerať na detaily a využívať pojmy zavedené v práci.

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Príloha A: obsah elektronickej prílohy

V elektronickej prílohe priloženej k práci sa nachádza zdrojový kód programu a súbory s výsledkami experimentov. Zdrojový kód je zverejnený aj na stránke <http://mojadresa.com/>.

Ak uznáte za vhodné, môžete tu aj podrobnejšie rozpísať obsah tejto prílohy, prípadne poskytnúť návod na inštaláciu programu. Alternatívou je tieto informácie zahrnúť do samotnej prílohy, alebo ich uviesť na oboch miestach.

Príloha B: Používateľská príručka

V tejto prílohe uvádzame používateľskú príručku k nášmu softvéru. Tu by ďalej pokračoval text príručky. V práci nie je potrebné uvádzať používateľskú príručku, pokiaľ je používanie softvéru intuitívne alebo ak výsledkom práce nie je ucelený softvér určený pre používateľov.

V prílohách môžete uviesť aj ďalšie materiály, ktoré by mohli pôsobiť rušivo v hlavnom texte, ako napríklad rozsiahle tabuľky a podobne. Materiály, ktoré sú príliš dlhé na ich tlač, odovzdajte len v electronickej prílohe.